

Saravanan Valliappan

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science in Computer Engineering | GPA: 4.0/4.0

Sep. 2025 – Jun. 2028

- Jacobs School Scholar, \$180,000 merit scholarship awarded to 20 out of 136,740 applicants

Relevant Coursework: *Computer Organization, Systems Programming, Analog Design, Circuits and Systems, Components & Circuits Lab*

University of California, Berkeley

Berkeley, CA

Summer Visitor | GPA: 4.0/4.0

Jun. 2024 – Aug. 2024

Relevant Coursework *Data Structures & Algorithms, Discrete Math*

EXPERIENCE

IEEE - Eta Kappa Nu

La Jolla, CA

Electrical Engineering Intern

Sep. 2025 - Present

- Collaborating with engineering team to develop microcontroller firmware in **C++** for real-time I/O handling (rotary encoders, tactile switches, SPI-based image transfer).
- Implementing interrupt-driven design to achieve sub-millisecond response latency for user input events.

International Tamil Academy

San Ramon, CA

Machine Learning Software Intern

Aug. 2023 - May 2025

- Designed and trained a **PyTorch CNN-based ML pipeline** for recognizing handwritten Tamil characters from unstructured image data; achieved 92% classification accuracy across 247 character classes.
- Built data preprocessing workflows using **NumPy, SciPy, and PIL**; developed **Flask-based application** to collect labeled input data, run inference, and visualize predictions for iterative model refinement.

FIRST Robotics Competition Team 7419

Dublin, CA

Fabrication Lead

Jan. 2023 – May 2025

- Operated **CNC router, CNC mill, laser cutter**, other industry-grade machining tools to fabricate robot parts.
- Used **CAD (Onshape, Fusion 360)** for design and prototyping; assist with electrical system layout and wiring.
- Designed and presented on how to effectively develop a robotics curriculum for students at the 2024 FIRST World Championships to over 100 students, mentors, parents, and teachers.

Vetri Labs

Fremont, CA

Battery Testing Intern

July 2024 – Dec. 2024

- Researched relationships between battery life and factors such as charge/discharge rate and chemistry.
- Built **Python-based data acquisition pipeline** interfacing with **Neware battery testing hardware**, processing test data to extract efficiency metrics using **pandas, NumPy, and Matplotlib**.
- Developed automated analysis algorithms reducing manual data processing time by 75%, enabling rapid iteration on cell chemistry evaluation and thermal management design optimization.

PROJECTS

AI-Powered File Search Engine | *Python, FastAPI, Next.js, ChromaDB, Ollama*

Dec. 2025

- Built a full-stack semantic search application enabling natural language queries across local documents using **vector embeddings** and **cosine similarity**, processing 1000+ files with < 1s query response time.
- Implemented async indexing pipeline with **FastAPI background tasks**, supporting PDF, DOCX, TXT with smart text chunking to preserve context.
- Designed **React/Next.js** frontend with file-level match visualization showing and relevance percentages.

SKILLS

Programming Languages: Java, Python, JavaScript, C, C++, Assembly, MATLAB, R, MySQL, HTML/CSS

Systems & Software Tools: UNIX/Linux, Git, SPICE, gdb, valgrind, CMake, Visual Studio Code, IntelliJ IDEA

Libraries & Frameworks: pandas, NumPy, Matplotlib, FastAPI, Next.js, Flask, Express.js

CAD & Fabrication Software: Onshape, Fusion 360, VCarve Pro, LightBurn, Ultimaker Cura, Bambu Studio

Fabrication Technologies: CNC Routing/Milling, Laser Cutting, FDM 3D Printing, Electrical Soldering